

Computer Science & IT

Discrete Mathematics



Set Theory & Algebra

Lecture No. 18



By- Vishal Sir

Recap of Previous Lecture



Topic

Functions



Topic

Domain / Range of a function



Topic

Injective (one-one) function



Topic

Surjective (onto) function



Topics to be Covered



Topic

Number of onto functions



Topic

Bijjective function

Topic

Inverse of a function

Topic

Identity function

Topic

Constant function

Topic

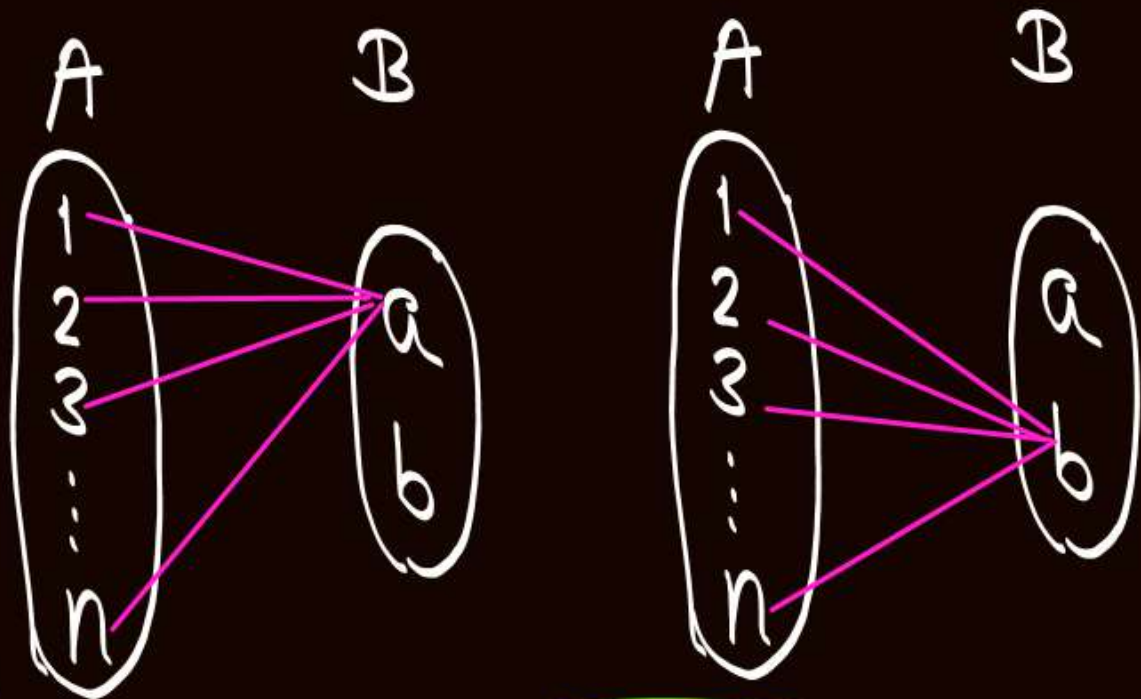
Identical function

Q:- If $|A|=n$ and $|B|=2$ (s.t. $n \geq 2$)
then How many onto functions are possible from A to B .

Total no. of onto
Functions from A to B

= Total no. of
Functions from
 A to B

No. of Functions
from A to B
that are not onto



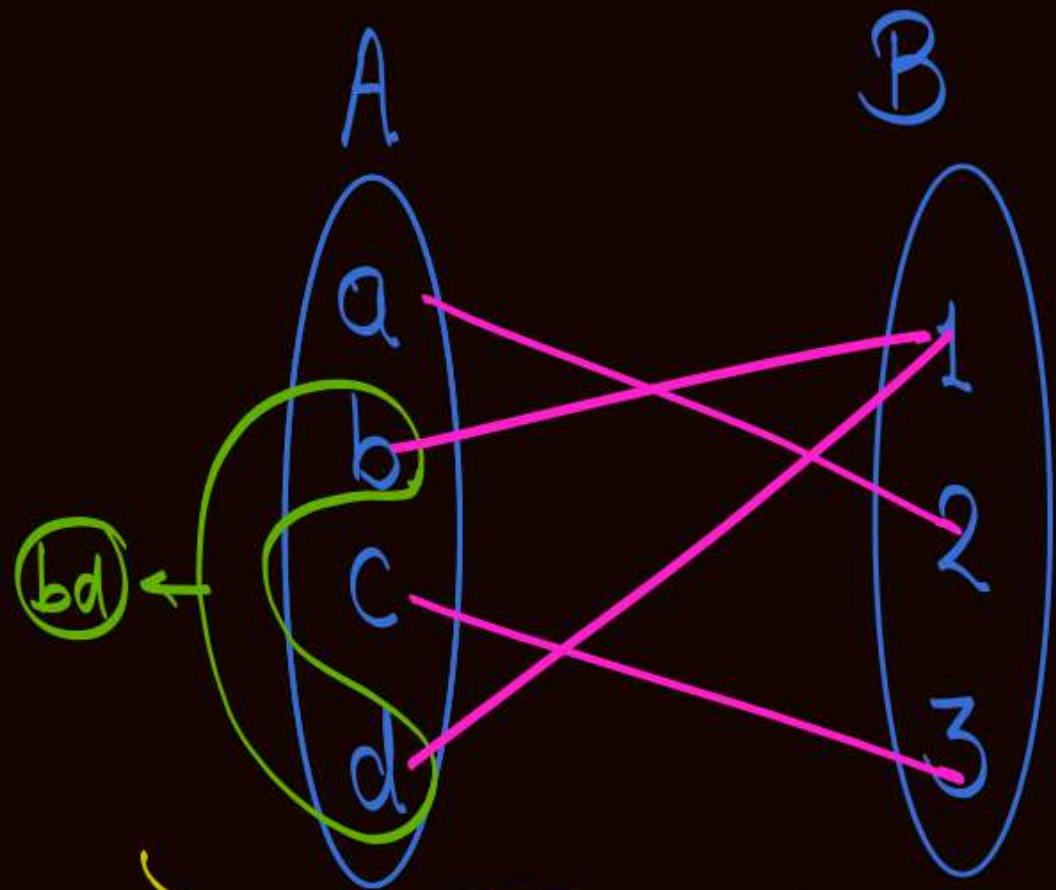
These are the only two functions from
 A to B that are not onto

$$= 2^n$$

$$= \boxed{2^n - 2} \text{ Ans.}$$

$$2$$

Q:- If $|A|=4$ and $|B|=3$
then How many onto functions are possible from A to B.



∴ No. of onto
functions possible
from set A to set B

$$= {}^4C_2 * 3!$$

$$= 6 * 6$$

$$= \underline{\underline{36}} \text{ Ans}$$

If we want to define any
Onto function from set A to set B
then exactly two elements of set A
will relate with the same element of set B

Q:- If $|A|=n$ and $|B|=n-1$ {i.e. Cardinality of domain = 1 + Cardinality of Co-domain} then How many onto functions are possible from A to B.

$$\therefore \text{No. of onto functions possible from set A to set B} = {}^nC_2 * (n-1)!$$

		No. of onto functions from set A to set B
①	$ A =n$ & $ B =2$ ($n \geq 2$)	$2^n - 2$
②	$ A =n$ & $ B =n$	$n!$
③	$ A =n$ & $ B =n-1$ $ A = 1 + B $	${}^nC_2 * (n-1)!$
④	$ A =m$ & $ B =n$ Where ($m \geq n$)	

Q: if $|A|=m$ & $|B|=n$ (such that $m \geq n$)
then how many onto functions are possible from set A to set B.

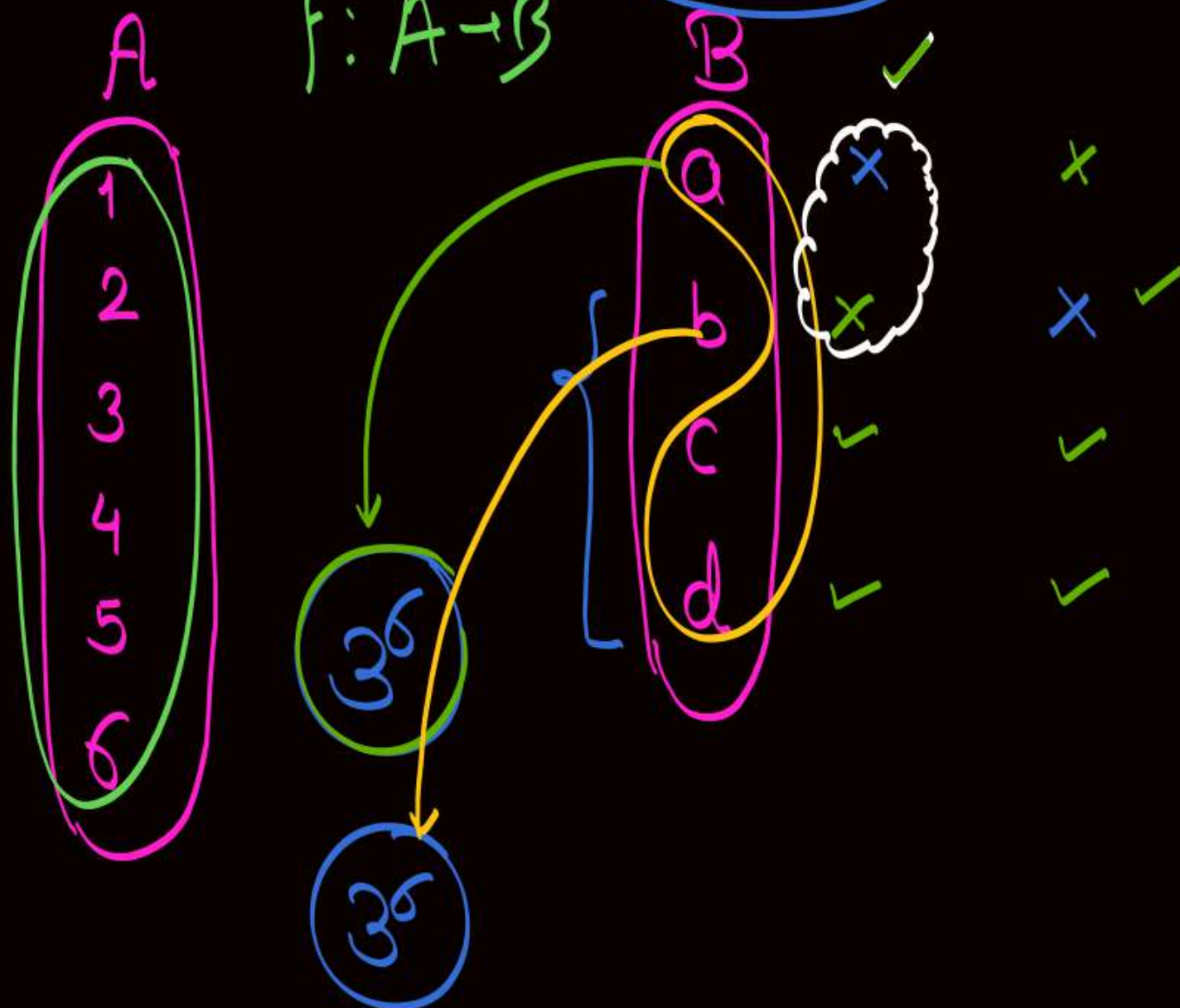
$$(n - (n-1)) = 1$$

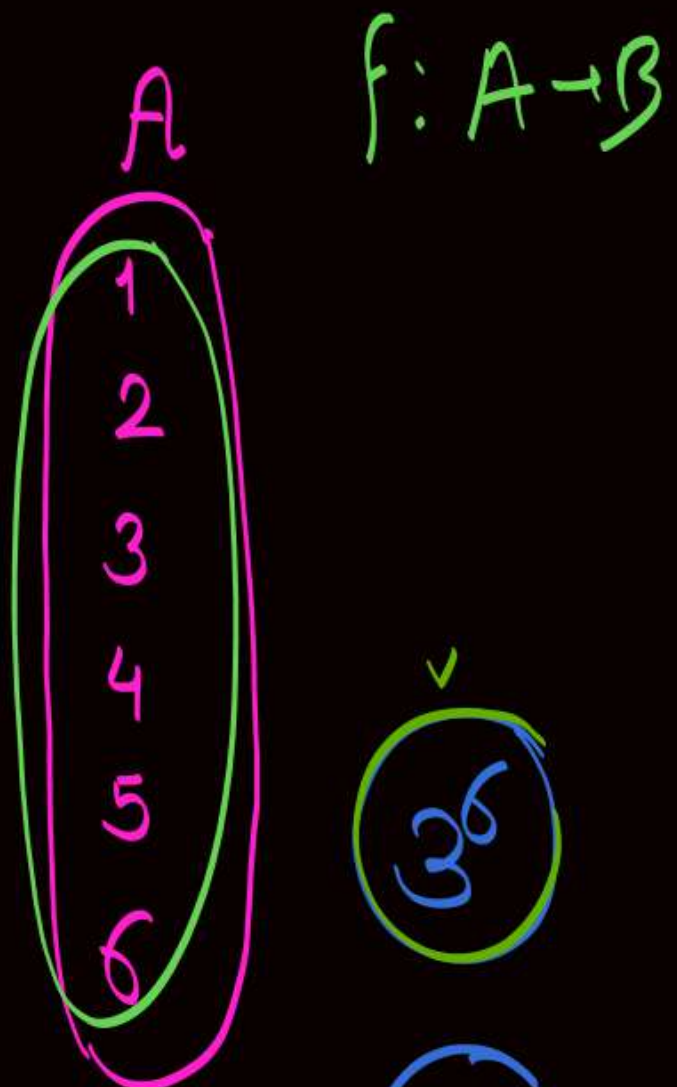
Total no. of
Onto functions
Possible from
Set A to set B

$$= n^m - {}^nC_1 (n-1)^m + {}^nC_2 (n-2)^m - {}^nC_3 (n-3)^m + {}^nC_4 (n-4)^m - {}^nC_5 (n-5)^m + \dots + (-1)^{(n-1)} {}^nC_{n-1} (1)^m$$

$$= {}^nC_0 (n-0)^m - {}^nC_1 (n-1)^m + {}^nC_2 (n-2)^m - {}^nC_3 (n-3)^m + \dots + (-1)^{n-1} {}^nC_{n-1} (n-(n-1))^m$$

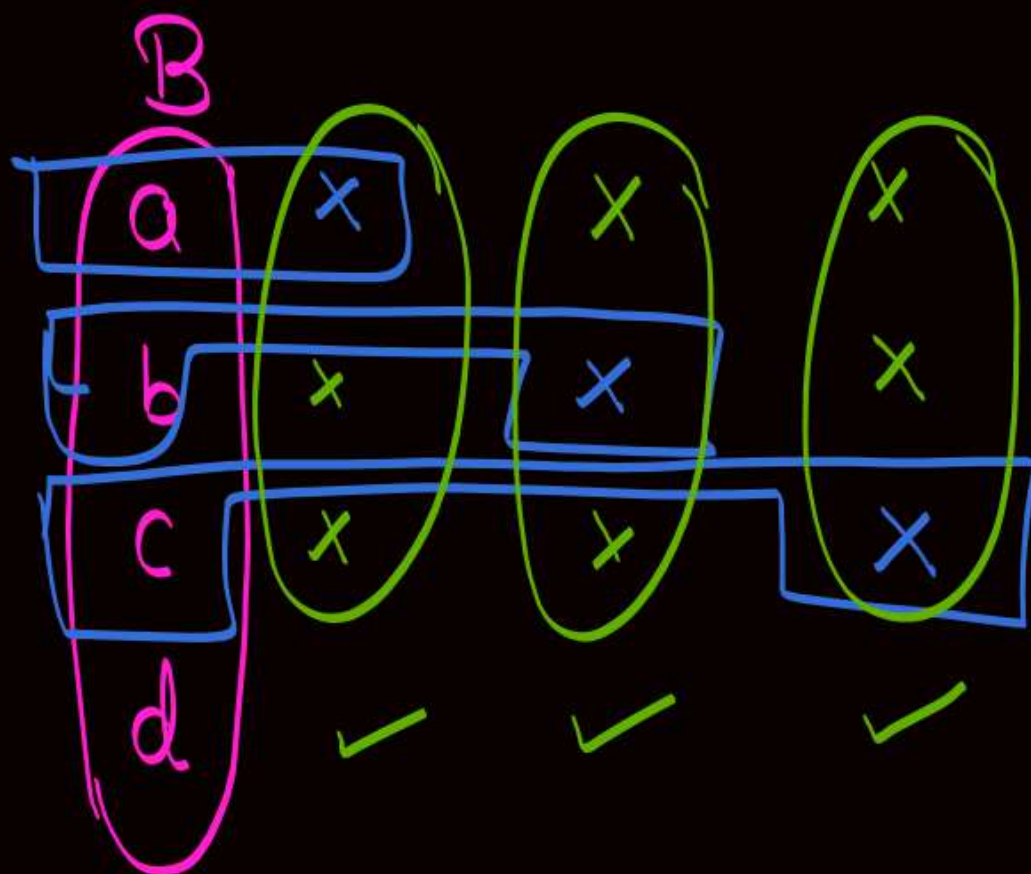
$$= \sum_{i=0}^{(n-1)} (-1)^i \cdot {}^nC_i \cdot (n-i)^m$$

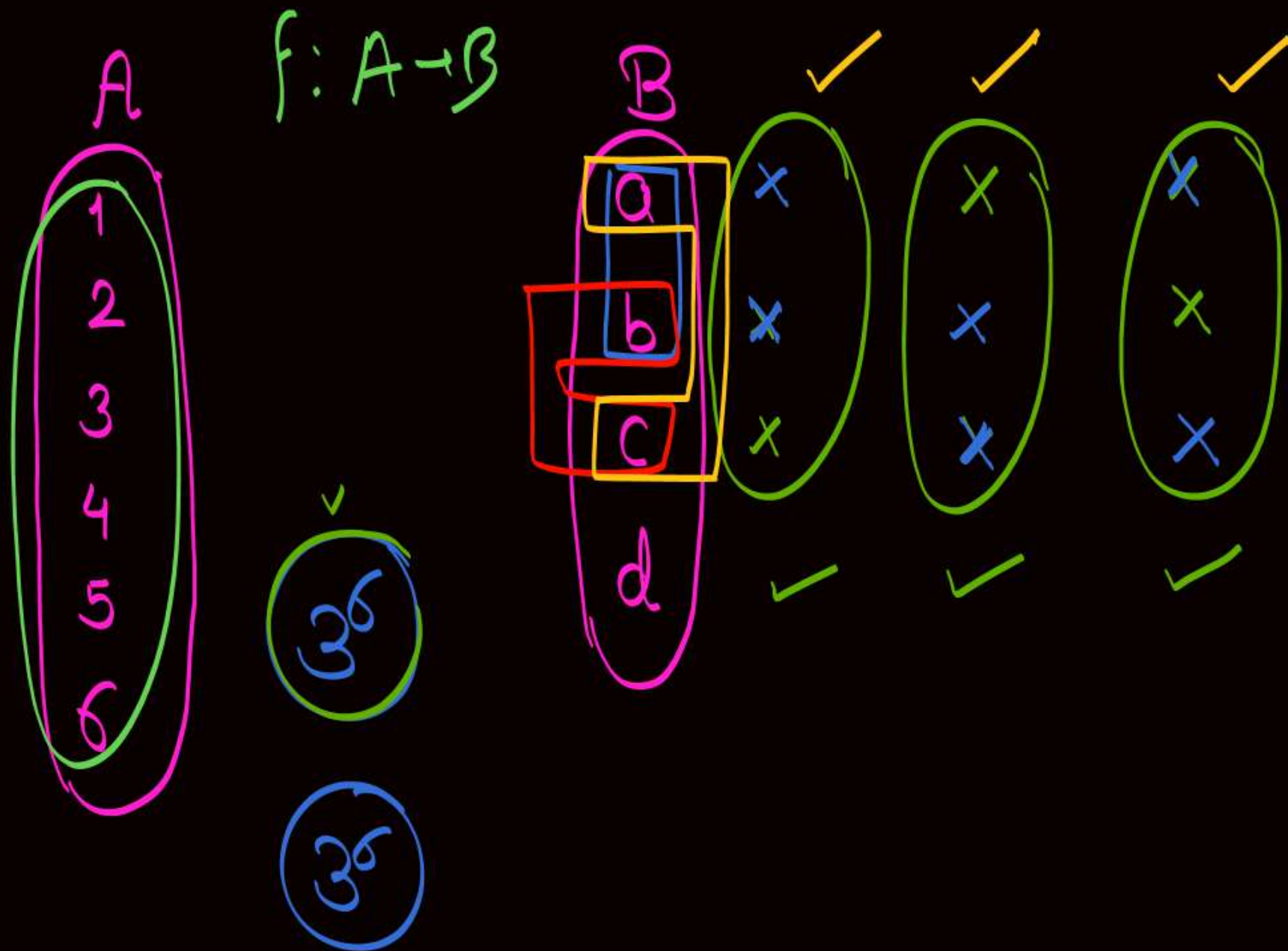
$$f: A \rightarrow B$$
$$|B| = n$$




36

36





		No. of onto functions from set A to set B
}	① $ A =n$ & $ B =2$ ($n \geq 2$)	$2^n - 2$
	② $ A =n$ & $ B =n$	$n!$
	③ $ A =n$ & $ B =n-1$ $ A = 1 + B $	$n \cdot 2^{n-2} \cdot (n-1)!$
	④ $ A =m$ & $ B =n$ Where ($m \geq n$)	$\sum_{i=0}^{n-1} (-1)^i \cdot nC_i \cdot (n-i)^m$



Topic : Bijective Function

A function $f: A \rightarrow B$ is called a bijective function if and only if

- ① $f: A \rightarrow B$ is one-one (injective)
- And ② $f: A \rightarrow B$ is onto (surjective)

* If A function from A to B is bijective then f is one-one and f is onto.
ie. $|A| \leq |B|$ and $|A| \geq |B|$

ie. A bijective function is possible from set A to set B if and only if $|A| = |B|$

* If $|A| = |B| = n$, then
Number of bijective functions
Possible from set A to set B $= n!$

Note :-

① If bijective function exist between the sets A and B ,
then Cardinality of set A = Cardinality of set B .

② If bijective function exist between the sets A and B ,
then set A and set B are said to have

One-one Correspondance.

(different from one-one function)

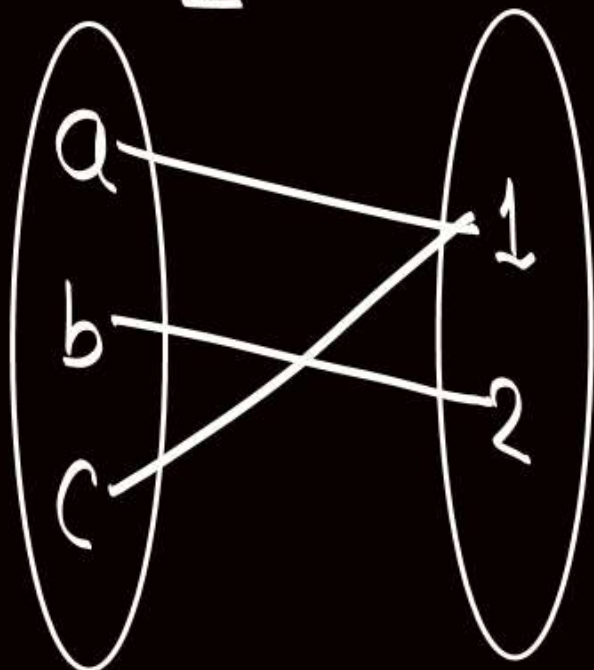


Topic : Inverse of a Function

- Inverse of a function "f" can exist if and only if, function "f" is a bijective function
- If $f: A \rightarrow B$ is a bijective function, then inverse of function 'f' can be denoted by " f^{-1} " and it will be defined from set B to set A

i.e $f: A \rightarrow B$ is invertible then $f^{-1}: B \rightarrow A$

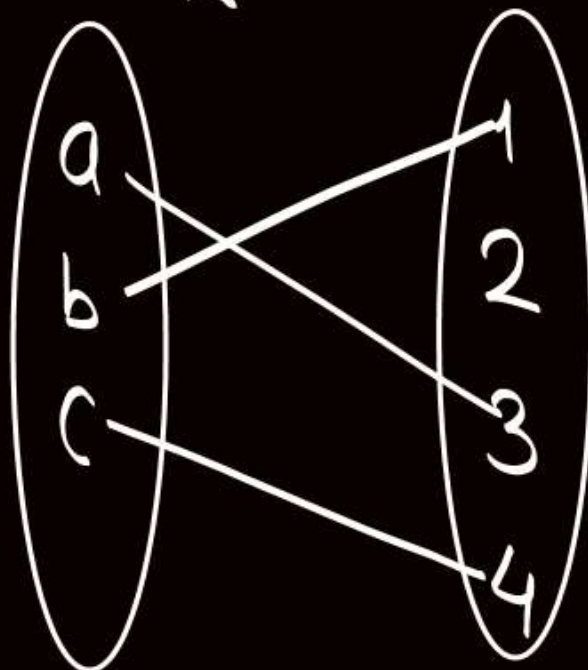
$A \quad f_1: A \rightarrow B \quad B$



f_1 is onto,
but not one-one

→ Inverse relation wrt.
function f_1 will not
be a function

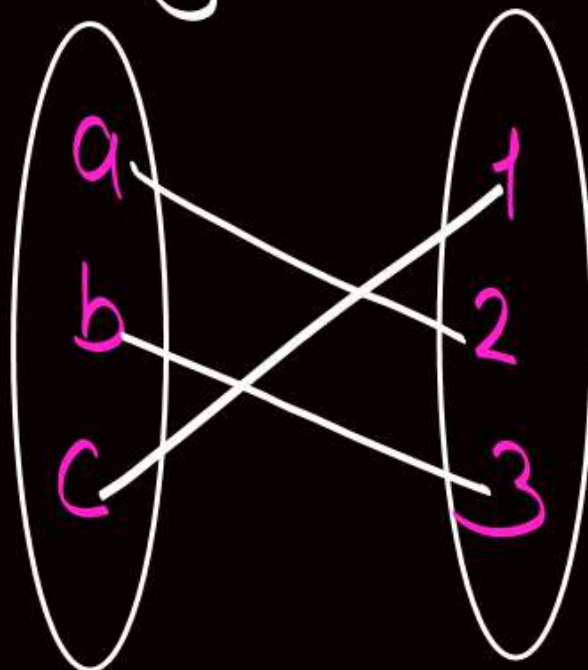
$A \quad f_2: A \rightarrow B \quad B$



f_2 is one-one
but not onto

→ Inverse relation wrt.
function f_2 will not
be a function

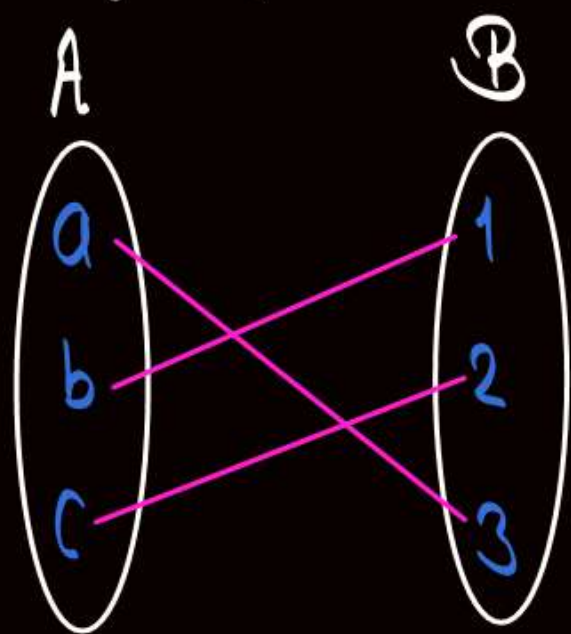
$A \quad f_3: A \rightarrow B \quad B$



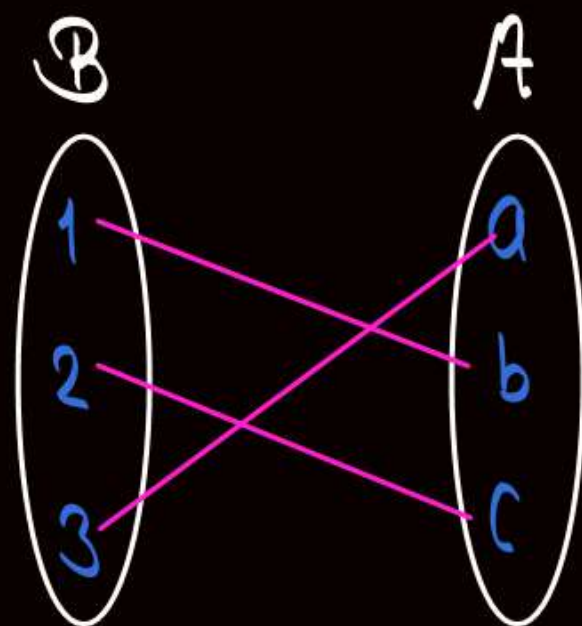
f_3 is one-one
as well as onto

→ Inverse relation wrt.
function f_3 , then it
will be a function

eg. let $f: A \rightarrow B$ is bijection $\therefore$



$f^{-1}: B \rightarrow A$



$$\begin{array}{ll} f(a) = 3 & \longrightarrow f^{-1}(3) = a \\ f(b) = 1 & \longrightarrow f^{-1}(1) = b \\ f(c) = 2 & \longrightarrow f^{-1}(2) = c \end{array}$$

i.e.,
if $f(x) = y$
then $f^{-1}(y) = x$

Q: Let 'f' is a bijective function
defined as $f(x) = 2x + 3$

Obtain the inverse function w.r.t. function 'f'.

i.e., $f^{-1}(x) = ?$

Soluⁿ:- let $f(x) = y \implies 2x + 3 = y$

$\therefore f^{-1}(y) = x$

└ $f^{-1}(y)$ must be a function of y
∴ We need to represent ' x ' in terms of y

We know, $2x + 3 = y$

$\therefore x = \frac{y-3}{2}$

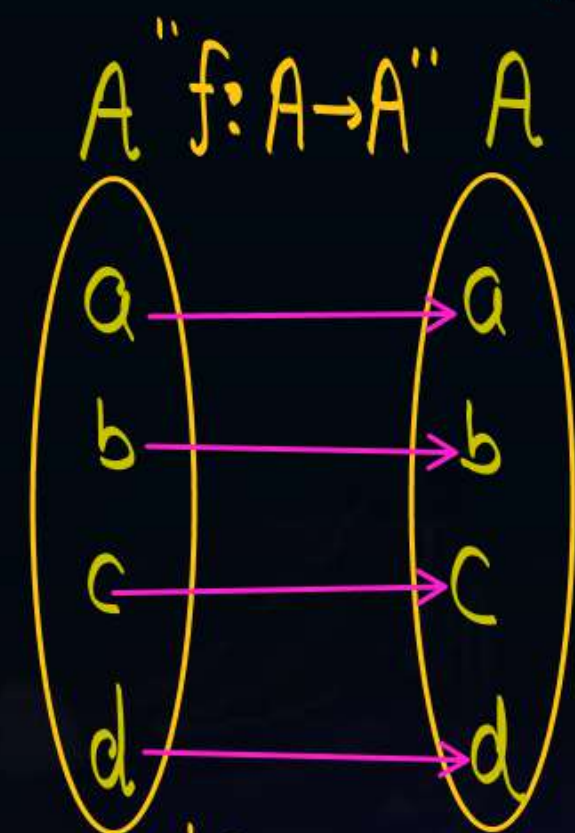
└ Hence $f^{-1}(y) = \frac{y-3}{2} \implies$ i.e., $f^{-1}(x) = \frac{x-3}{2}$



Topic : Identity Function

A function "f" from set A to set A (i.e. on set A) is called an identity function if and only if $f(x) = x, \forall x \in A$

- Generally, Identity function on set A is denoted by I_A .
- ⊛ Every identity function is always a bijection
- * Inverse of identity function will always be that identity function itself i.e. $I_A^{-1} = I_A$



Here 'f' is an identity function

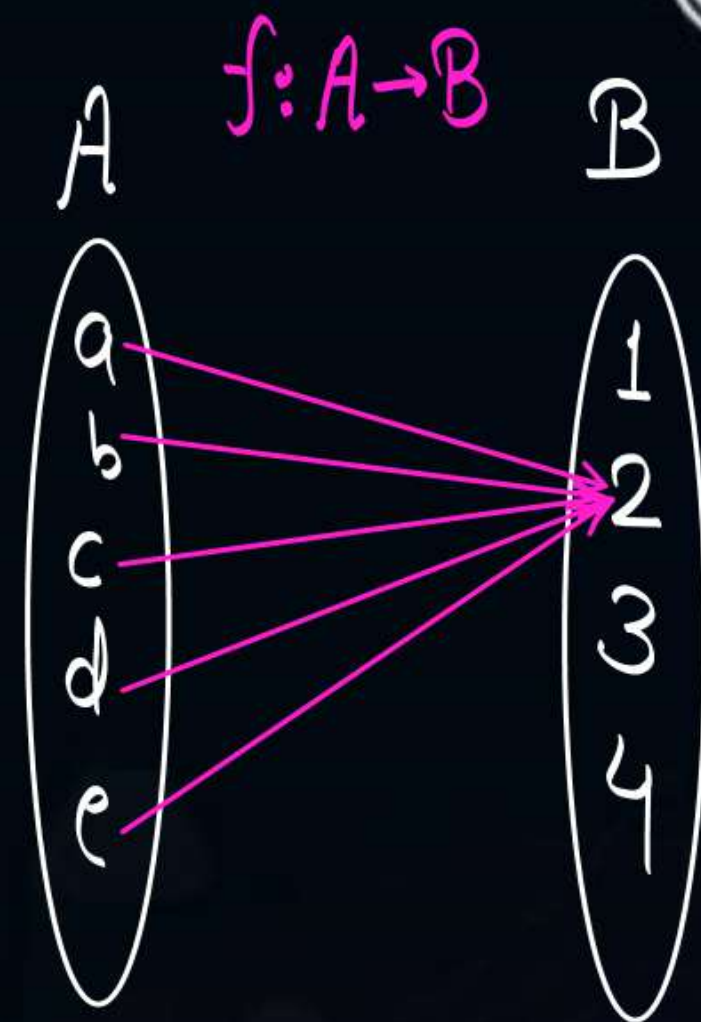


Topic : Constant Function

A function $f: A \rightarrow B$ is called a Constant function, if and only if, $f(x) = c, \forall x \in A$

$\swarrow$
A fixed value from set B

• If $|A| = m$ and $|B| = n$, then
No. of Constant functions possible from set A to set B $= n = \text{Cardinality of Co-domain}$



$f(x) = 2, \forall x \in A$

$\swarrow$
Constant



2 mins Summary



✓ Topic

Number of onto functions

✓ Topic

Bijjective function

✓ Topic

Inverse of a function

✓ Topic

Identity function

✓ Topic

Constant function

Topic

Identical function

THANK - YOU